

How Many More Labels Are Needed? Minimax Evaluation under Selective Labels

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More labels help only when the audit experiment supplies identifiable information.

RESEARCH QUESTION

How many unresolved labels must be audited to detect excessive risk while controlling false alarms under outcome-dependent missingness?

57	94 → 367	2,645
Exact finite minimum in the declared truthful design	Numerical limiting budgets: known 10% channel → conditional bounds	Minimum complete-label cohort for 10% versus 12% risk

Problem and protocol

A frozen cohort can look accurate because errors remain unlabeled. The withholding process may inspect every outcome, but must commit a truthful partial ledger before private queries begin. The evaluator has a hard query budget and one decision deadline.

Exact benchmark

The paper characterizes fixed-query asymptotic minimax power over full-record adaptive policies. Uniform nonadaptive truthful queries attain the frontier with finite-sample false-alarm control. The frontier depends on missingness relative to the risk gap.

Finite design certificate

At 1,000 predictions, 500 pending labels, 10% versus 20% risk, and a 5% false-alarm ceiling, exactly 57 truthful queries are necessary and sufficient for 95% worst-case power under the count-tail classes. Budget minimality does not identify exact optimal power.

Reliability information has a cost

For 95% limiting power at the same risks and missing fraction, a known 10% symmetric noise channel needs 94 queries numerically; only knowing the corresponding conditional reliability bounds needs 367. At 20% bounds, every fixed budget lies in an asymptotic blind region.

Smaller violations need more information

The frozen 57-query rule has only 1.15% power at 12% risk. A private mixture raises this to 8.66% while preserving at least 95.017% at 20% risk. Even complete labels at $n = 1,000$ have only 65.32% optimal power for 10% versus 12%; the first complete-label cohort reaching 95% is $n = 2,645$.

EVIDENCE AT A GLANCE

Resource	What can fail without it
Enough queries	Hidden error compositions remain difficult to distinguish.
Reliable label information	Different channels can produce identical responses.
Enough predictions	Even complete labels have insufficient power.

TAKEAWAY

Plan label count, label-quality validation, and cohort size jointly. A nonrejection is not a certificate of acceptable risk.

INTERPRETATION & LIMITS

Mathematical, single-deadline experiment with accessible existing outcomes. Risk classes and conditional label-quality bounds require external justification. Finite certificates and asymptotic numerical budgets have different scope. No deployment validation is claimed.